

- b. What is the minimum number of packets (window size W) that must be kept in flight to achieve 100% utilization?

Q2. Consider a reliable data transmission scenario between two hosts, A (sender) and B (receiver). Host A has **12 packets** to transmit and employs a Go-Back-N protocol with **window size $N=4$** and sequence numbers ranging from 0 to 15. If acknowledgements sent by host B were never lost, and only the **3rd packet (having sequence number 2)** was lost, how many packets in total were transmitted by host A? Support your answer through a timing diagram. **[10 Marks] [CLO 2]**